

PROJECT PROPOSAL

Resource Coordination Platform for Community Resilience

GROUP NO: 14

PID: 09

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1. Title of the Project

Resource Coordination Platform for Community Resilience

2. Overview of the Project

The Resource Coordination Platform for Community Resilience is a digital platform designed to improve the coordination of resources, volunteers, and relief activities during emergencies and community crises such as floods, landslides and other disaster situations. The system provides a centralized platform where community members can request assistance, donors can offer resources, volunteers can participate in relief activities and coordinators can manage requests, inventory, resource allocation and volunteer assignments through a unified dashboard [1], [2], [3]. The proposed solution consists of a responsive web application for coordinators and administrators and a cross-platform mobile application for community members, donors, and volunteers.

3. Objectives of the Project

The objectives of this project are to:

- Design and implement a centralized platform for coordinating community resources and relief operations during emergencies [1], [2].
- Provide secure and efficient management of help requests, donations, inventory, and volunteer activities.
- Develop tools for coordinators to verify requests, allocate resources, assign volunteers, and monitor operations.
- Improve the efficiency, transparency, and reliability of community-based emergency response efforts.

4. The Need for the Project

Community organizations often coordinate relief efforts using phone calls, messaging applications, social media, and manual records, leading to duplicated requests, inefficient resource allocation and poor volunteer coordination. A centralized platform is needed to manage requests, donations, inventory and volunteer activities while ensuring efficient resource distribution, improved coordination and protection of sensitive community information during emergencies.

5. Scope of the Project

The system will support community members, donors, volunteers and coordinators through functionalities including user authentication, help request management, donation management, inventory tracking, volunteer registration and assignment, task management and administrative reporting. The platform will support priority based resource allocation, request verification, real time notifications, multi-tenant organization support, offline data synchronization and operation under low bandwidth conditions.

6. Deliverables

- A responsive web application primarily designed for coordinators and administrators to manage emergency response operations.
- A cross platform mobile application for community members, donors, and volunteers to request assistance, offer resources, and participate in relief activities.
- A secure backend system with APIs, database management, real time communication, and role based access control.
- Comprehensive project documentation including requirements, UML diagrams, database design, testing reports, and user documentation.

7. Overview of Existing Systems and Technology

Existing disaster management and community coordination platforms such as Sahana Eden, Ushahidi and Crisis Cleanup provide functionalities including resource management, volunteer coordination, and incident reporting [4], [5], [6]. However, these systems are primarily designed for large organizations and often lack a lightweight, community focused approach suitable for grassroots initiatives.

Proposed Technologies:

React.js, Node.js with Express.js, PostgreSQL, JWT Authentication, Socket.IO, Flutter, Docker, Git/GitHub, and GitHub Projects.

8. References

- [1] Betke, H., et al., "A Design Theory for Spontaneous Volunteer Coordination Systems in Disaster Response," Proceedings of the Hawaii International Conference on System Sciences, 2024. Available: <https://scholarspace.manoa.hawaii.edu/items/6e1f845c-7e16-492b-b48d-c4a6bd4f09e5> (Accessed: 02-Jul-2026)
- [2] D. A. Carpenter et al., "Resource exchange patterns between Voluntary Organizations Active in Disaster (VOADs): A multilevel network assessment to improve disaster response capacity," International Journal of Disaster Risk Reduction, vol. 108, 2024.
- [3] M. Sperlinga ., "Decision support for disaster relief: Coordinating spontaneous volunteers," European Journal of Operational Research, vol. 299, no. 2, pp. 690-705, 2022
- [4] Ushahidi, "Crowdsourcing Solutions to Empower Communities." Available: <https://www.ushahidi.com/> (Accessed: 01-Jul-2026).
- [5] Sahana Software Foundation, "Sahana Eden Disaster Management Platform." Available: <https://sahanafoundation.org/eden/> (Accessed: 01-Jul-2026).
- [6] Crisis Cleanup, "Crisis Cleanup Platform." Available: <https://api.crisiscleanup.org/> (Accessed: 01-Jul-2026).